COURSE NAME

DISCRETE MATHEMATICS

CSC 1204

(UNDERGRADUATE)

CHAPTER 6

GRAPHS

M. MAHMUDUL HASAN

ASSISTANT PROFESSOR, CS, AIUB

http://www.dit.hua.gr/~m.hasan

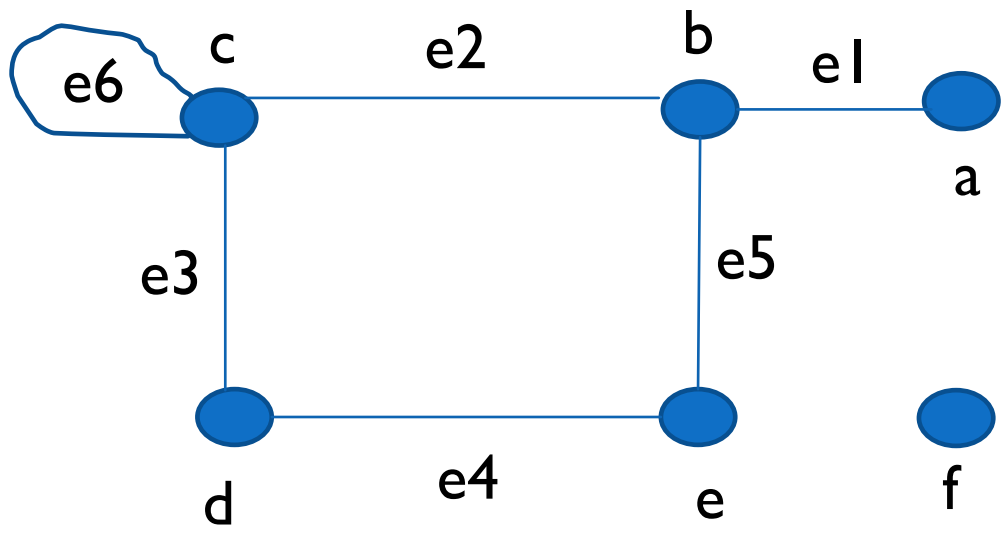






GRAPH

- A graph G = (V,E) consists of V, is a nonempty set of vertices (or nodes) and E, is a set of edges.
- Each edge has either one or two vertices associated with it, called its *end points*. An edge is said to *connect* its endpoints.
- **Note**: The set of vertices V of a graph may be infinite. A graph with an infinite vertex set is called an **infinite graph**. A graph with a finite vertex set is called a **finite graph**. [We will consider only finite graphs]



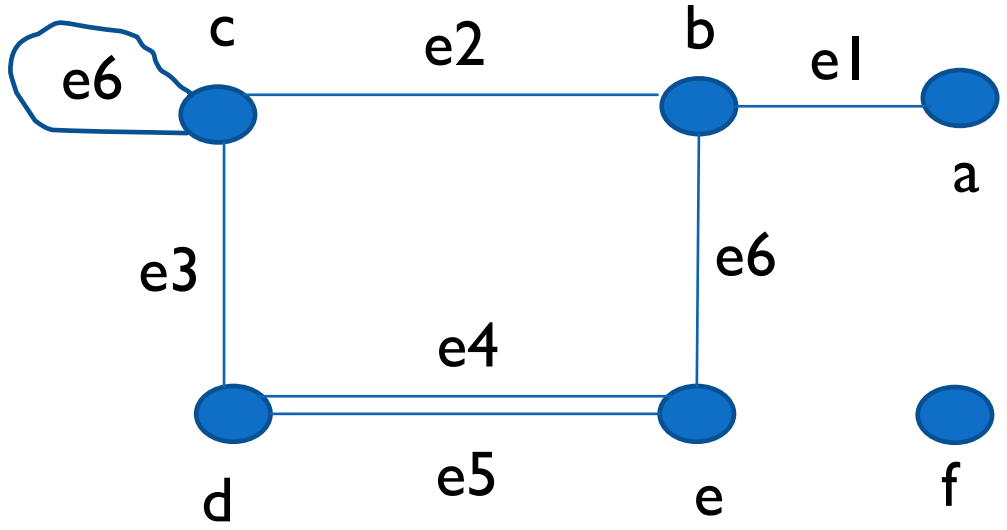
G = (V, E) where

- G is the name of the graph
- V is the set of vertices of the graph G
- E is the set of edge in the graph G

V = {a, b, c, d, e, f}
E = {e1, e2, e3, e4, e5, e6}

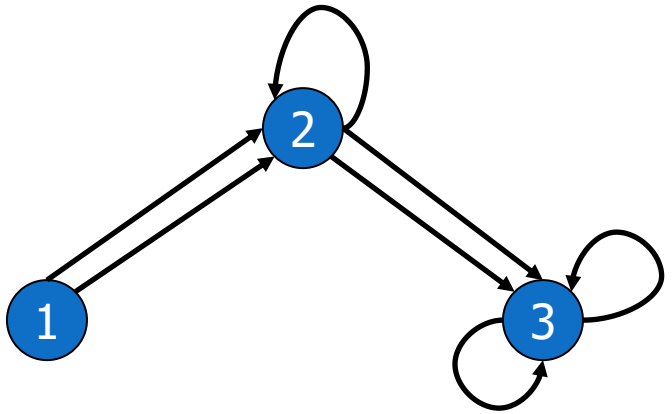
UNDIRECTED GRAPH

- Simple Graph: An *undirected* graph with no multiple edges or loops is called a simple graph.
- Multigraph: An *undirected* graph that may contain multiple edges connecting the same vertices but no loops.
- Pseudograph: An *undirected* graph that may contain multiple edges and loops is called a pseudograph.



DIRECTED GRAPH

- A directed graph(or digraph) (*V*,*E*) consists of a nonempty set of vertices *V* and a set of directed edges *E*.
- Each directed edge is associated with an ordered pair of vertices.
- The directed edge associated with the ordered pair (*u*,*v*) is said to *start* at ***u*** and *end* at ***v***.
- Simple Directed graph: When a *directed* graph has no loops and has no multiple directed edges, it is called a simple directed graph.
- Directed multigraph: A graph with *directed edges* that may contain multiple directed edges is called a directed multigraph.
- Mixed Graph: A graph with *both directed and undirected edges* is called a mixed graph. A mixed graph may contain loop(s).
- Loop: An edge that connect a vertex to itself is called a loop.

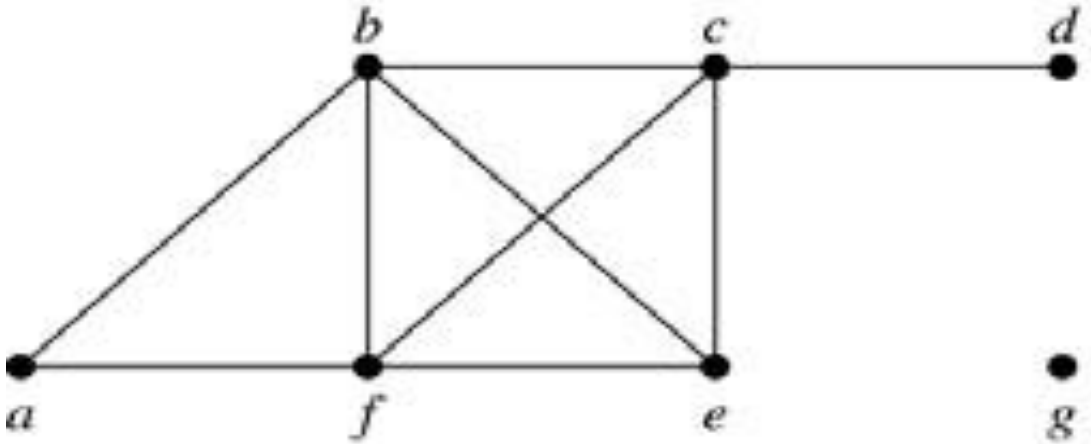


DIRECTED GRAPH

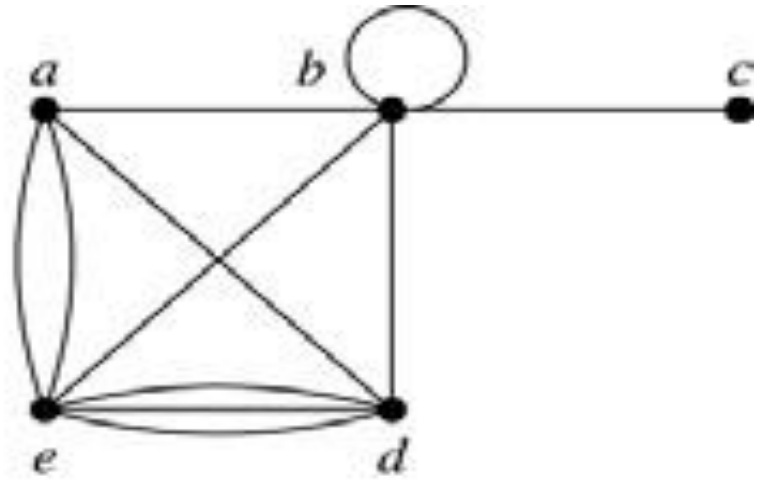
- Adjacent Vertices: Two vertices *u* and *v* in an undirected graph *G* are called *adjacent* (or *neighbors*) in *G* if *u* and *v* are endpoints of an edge of *G*.
- Incident: If e is associated with {*u*, *v*}, the edge e is called *incident* with the vertices *u* and *v*. The edge e is also said to *connect* *u* and *v*.
- Endpoints: The vertices *u* and *v* are called *endpoints* of an edge associated with {*u*, *v*}.
- Degree of vertices: The *degree of a vertex in an undirected graph* is the *number of edges incident with it*, except that a *loop* at a vertex contributes *twice* to the degree of that vertex. The degree of the vertex *v* is denoted by deg(*v*)
- Isolated vertex: A vertex of degree zero is called isolated.
- Pendant vertex: A vertex is pendant if and only if it has degree one.

DEGREE OF VERTICES

Q: What are the degrees of the vertices in the graphs G and H?



G



H

Solution:

G: deg(*a*) = 2, deg(*b*)= 4, deg(*c*)= 4, deg(*d*)=1, deg(e)= 3, deg(*f*)=4, and deg(*g*)= 0

H: deg(*a*) = 4, deg(*b*) = deg(e) = 6, deg(*c*) = 1, deg(*d*) = 5.

THE HANDSHAKING THEOREM

Theorem 1: Let G = (V,E) be an ***undirected graph*** with **e** edges. Then $2e = \sum_{v \in V} \deg(v)$

Q: How many edges are there in a graph with 10 vertices each of degree six?

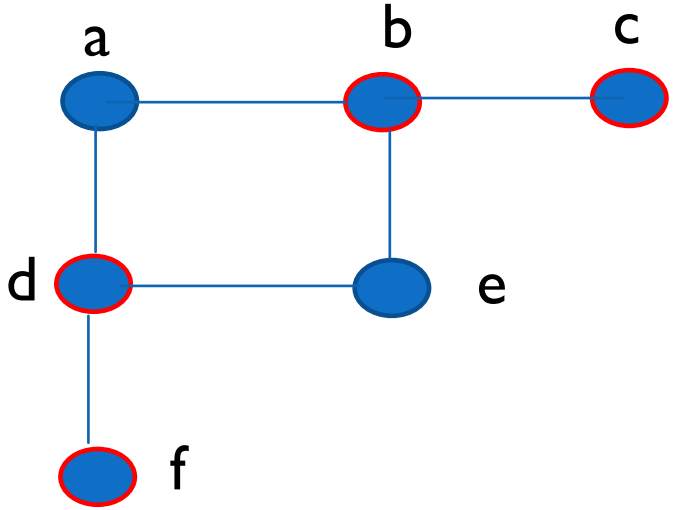
Solution: Because the sum of the degrees of the vertices is 6 x10 = 60, it follows that 2e=60. Therefore, e = 30

Theorem 2: An undirected graph has an even number of vertices of odd degree.

Q: If a graph has 5 vertices, can each vertex have degree 3?

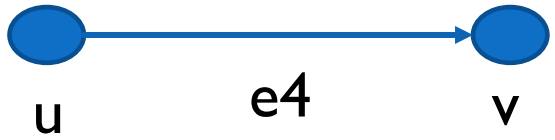
Solution:

- This is not possible by the Handshaking theorem, because the sum of the degrees of the vertices 3 x 5 = 15 is odd. (i.e. total edge = 15/2 = 7.5 which is a fraction)
- In the graph the odd degrees are: deg(b) = 3, deg(c) = 1, deg(d) = 3, deg(f) = 1. Here, the number of vertices that has odd degree are even which is 4 (e.g. b, c, d, f)



INITIAL VERTEX & TERMINAL VERTEX

- When (u, v) is an edge of the graph G with directed edges, *u* is said to be *adjacent to v* and *v is said to be adjacent from u*.
- The vertex u is called the *initial vertex* of (u, v) and v is called the *terminal/end vertex* of (u, v).
- The initial vertex and terminal vertex of a loop are the same.



In-degree & Out-degree of a vertex

- In a graph with directed edges the *in-degree of a vertex v*, denoted by $\deg^-(v)$, is the number of edges with v as their terminal vertex. The *out-degree of v*, denoted by $\deg^+(v)$, is the number of edges with v as their initial vertex.
- Note: A loop at a vertex contributes 1 to both the in-degree and the out-degree of this vertex.

IN-DEGREE & OUT-DEGREE OF A VERTEX

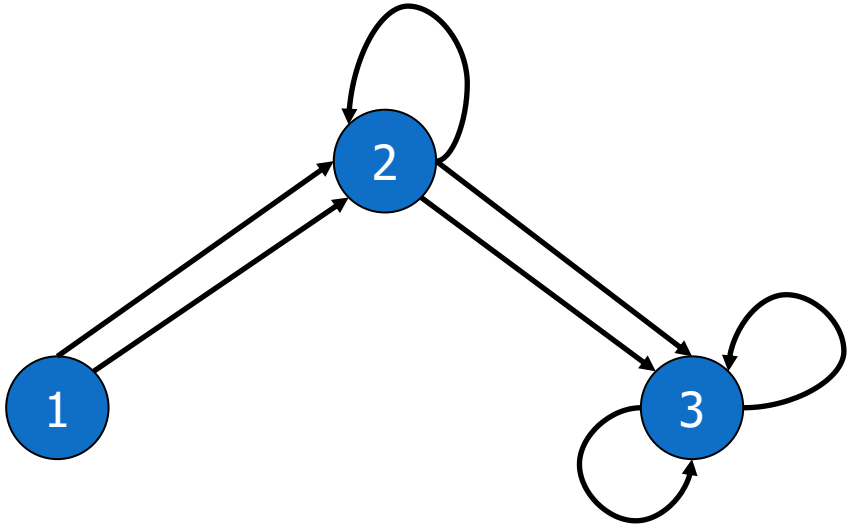
Q: What are in-degrees and out-degrees of all the vertices in the graph below?

Solution:

$\deg^-(1) = 0$

$\deg^-(2) = 3$

$\deg^-(3) = 4$



$\deg^+(1) = 2$

$\deg^+(2) = 3$

$\deg^+(3) = 2$

Handshaking Theorem 3 for directed graph: Let G = (V, E) be a graph with *directed edges*. Then

$$\sum_{v \in V} \deg^-(v) = \sum_{v \in V} \deg^+(v) = |E|$$

| E | is total number of Edge in the graph

SIMPLE GRAPHS: COMPLETE GRAPH (K_N)

- The complete graph on *n* vertices, denoted by K_n, is the simple graph that contains exactly one edge between each pair of distinct vertices.
- The graph , for n = 1, 2, 3, 4, 5, 6 are displayed in the following figure:

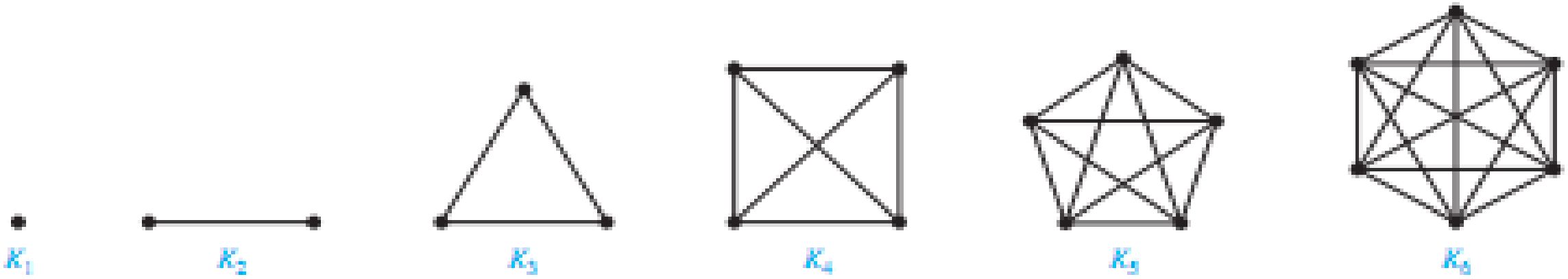


FIGURE 3 The Graphs K_n for $1 \leq n \leq 6$.

SIMPLE GRAPHS: CYCLES(C_N)

- The Cycle C_n, n>=3, consists of n vertices $v_1, v_2, \ldots v_n$ and edges $\{v_1, v_2\}, \{v_2, v_3\}, \ldots \{v_{n-1}, v_n\}$, and $\{v_n, v_1\}$
- The cycles for C_3, C_4, C_5, and C_6 are displayed in the following figure:

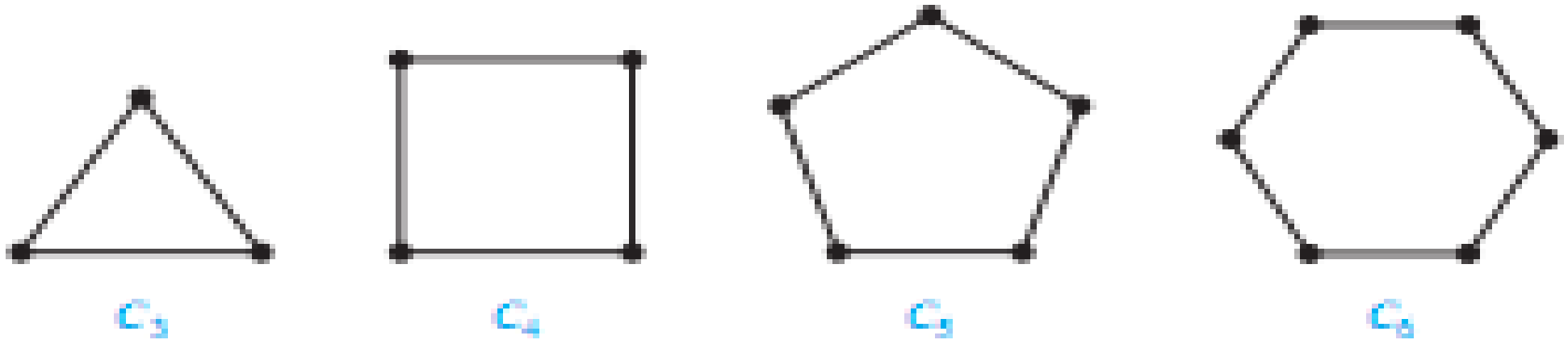
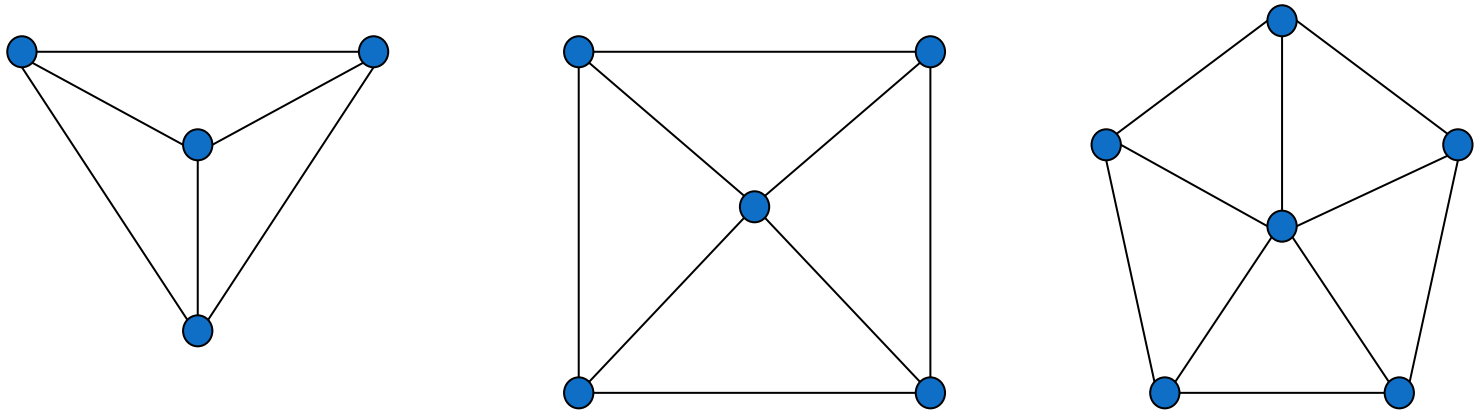


FIGURE 4 The Cycles C_3, C_4, C_5, and C_6.

SIMPLE GRAPHS: WHEELS(W_N)

- We obtain the Wheel (W_n) when we add an additional vertex to the Cycle C_n , for n>=3, and connect this new vertex to each of the n vertices in C_n , by new edges.
- The *wheel* W_n is just a cycle graph with an extra vertex in the middle
- The Wheels W_3 , W_4 , W_5 *are displayed in the figure below:*



SIMPLE GRAPHS: *N*-CUBES(Q_N)

- The *n*-dimensional hypercube, or *n*-cube, denoted by Q_n, is the graph that has vertices representing the 2^n bit strings of length *n*.
- Two vertices are adjacent if the bit strings that they represent differ in exactly one bit position.
- The graphs Q_1, Q_2 , and Q_3 are displayed in the following figure:

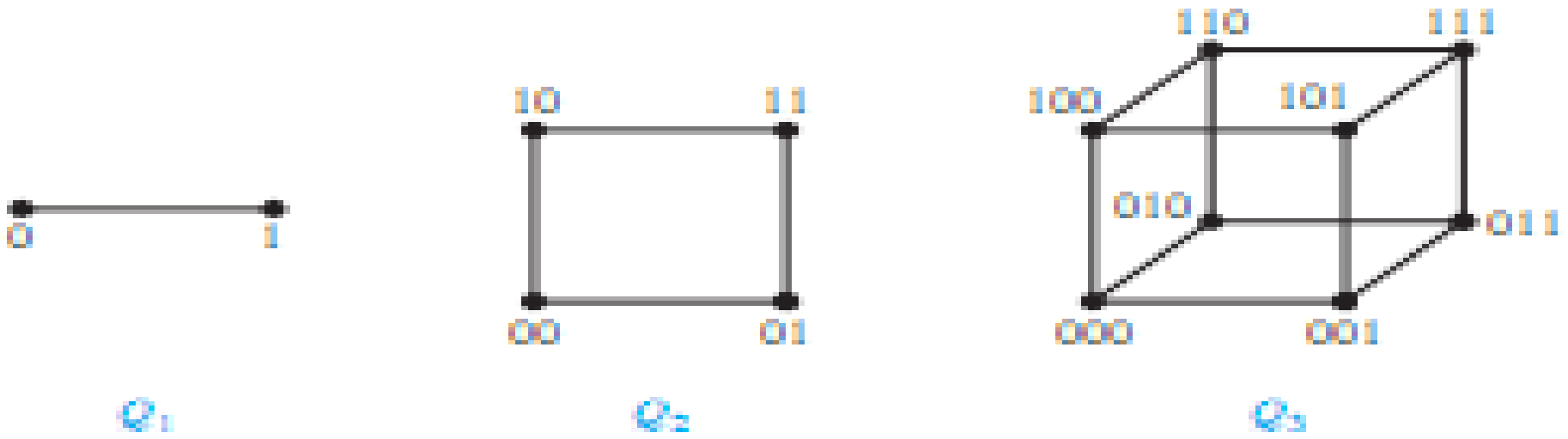


FIGURE 6 The *n*-cube Q_n, $n = 1, 2, 3$.

SIMPLE GRAPH: BIPARTITE GRAPHS

- A simple graph G is called ***bipartite*** if its vertex set V can be partitioned into two disjoint sets V_1 and V_2 such that every edge in the graph connects a vertex in V_1 and a vertex in V_2
- Note: If each vertex of V_1 is connected to each vertex of V_2, then it is called complete bipartite graph and it is denoted by $K_{m,n}$ where m is the number of vertices in V_1 and n is the number of vertices in V_2

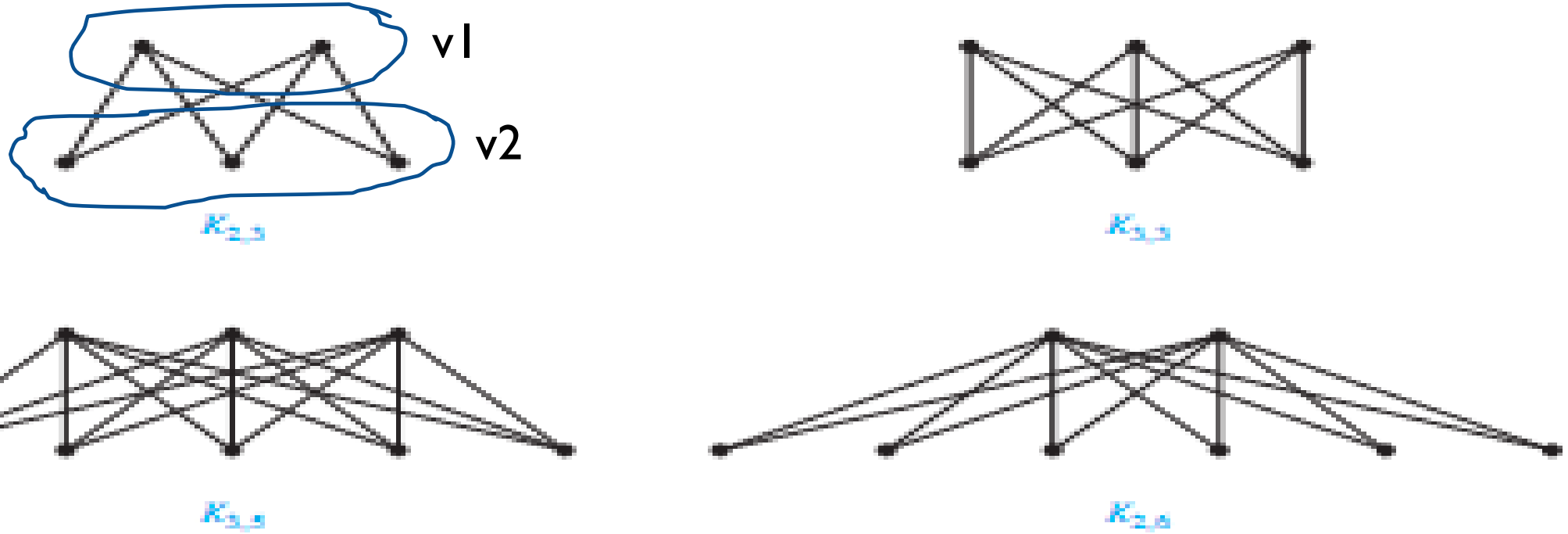
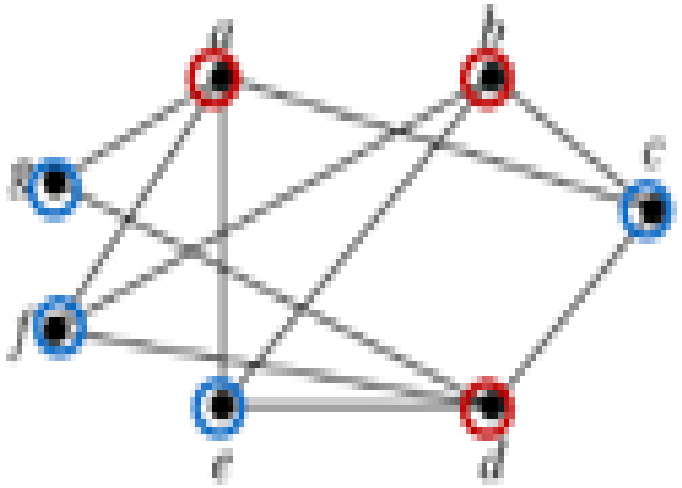


FIGURE 9 Some Complete Bipartite Graphs.

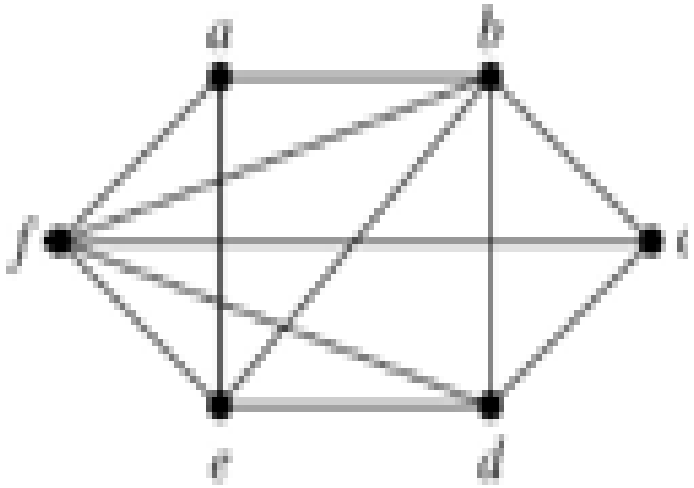
SIMPLE GRAPH: BIPARTITE GRAPHS

- An equivalent definition of bipartite graph is one where it is possible to color the vertices either red or blue so that no two adjacent vertices are the same color.

Q: Are The Graphs G And H Are Bipartite?



G



H

- G is bipartite
- H is not bipartite: if we color a red then its neighbors f and b must be blue, but f and b are also adjacent.

REPRESENTING GRAPHS: ADJACENCY LISTS

- One way to represent a graph without multiple edges is to **list** all the edges of the graph.
- Another way to represent a graph with no multiple edges is to use **adjacency lists**, which specify the vertices that are adjacent to each vertex of the graph.
- **Adjacency Lists**: A table with 1 row per vertex, listing its adjacent vertices.
- **Directed Adjacency Lists**: A table with 1 row per node, listing the *terminal nodes* of each edge *incident from that node*.

EX: Use adjacency lists to describe the simple graph given in Figure 1.

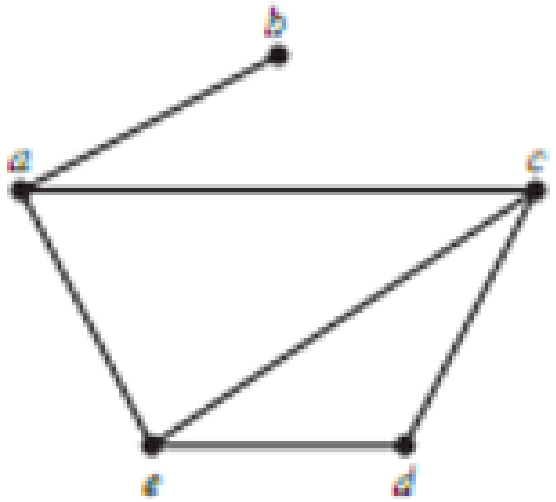


FIGURE 1 A Simple Graph

TABLE 1 An Adjacency List for a Simple Graph.

Vertex	Adjacent Vertices
a	b, c, e
b	a
c	a, d, e
d	c, e
e	a, c, d

REPRESENTING GRAPHS: DIRECTED ADJACENCY LISTS

EX: Represent the directed graph shown in Figure 2 by listing all the vertices that are the terminal vertices of edges starting at each vertex of the graph.

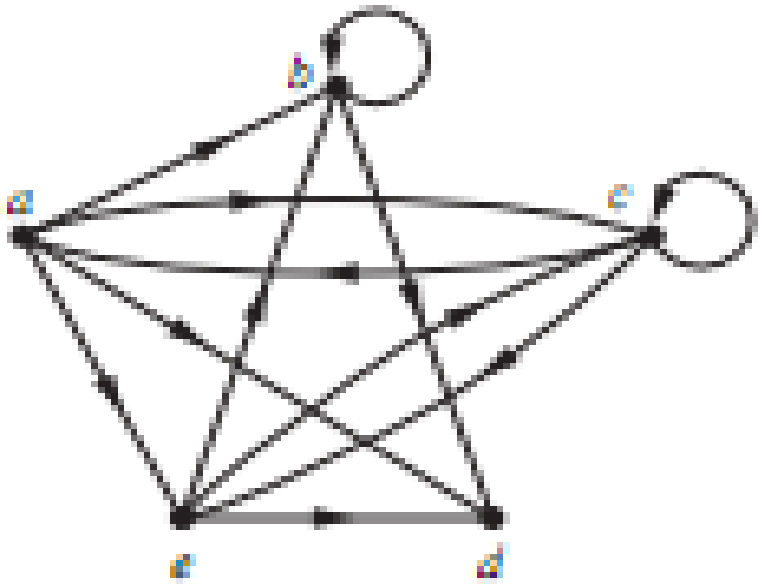


FIGURE 2 A Directed Graph.

TABLE 2 An Adjacency List for a Directed Graph.

Initial Vertex	Terminal Vertices
a	b, c, d, e
b	b, d
c	a, c, e
d	
e	b, c, d

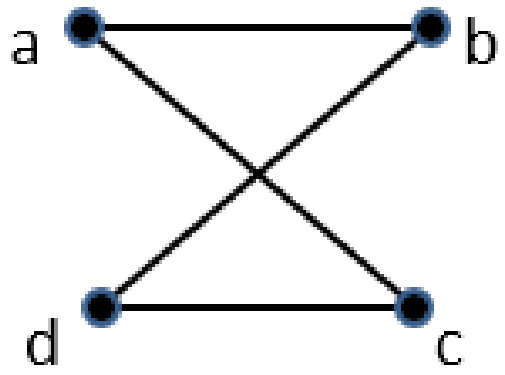
REPRESENTING GRAPHS: ADJACENCY MATRICES

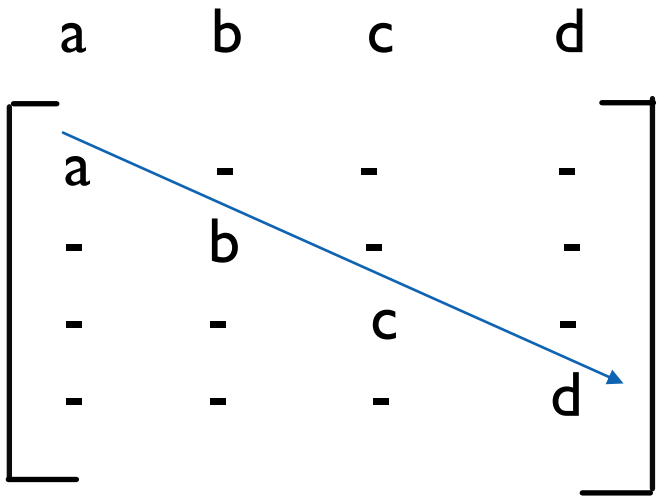
- Two types of matrices commonly used to represent graphs – Adjacency matrix, Incidence matrix
- Adjacency matrix: A matrix representing a graph using the adjacency of vertices.
- Incidence matrix: A matrix representing a graph using the incidence of edges and vertices.

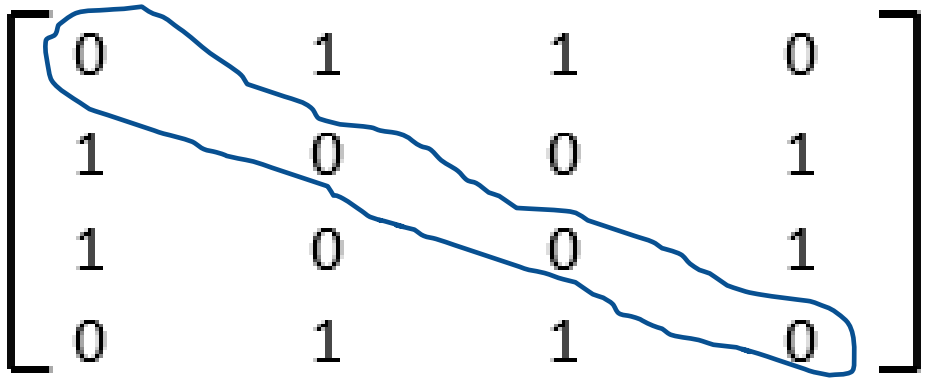
Q: Draw a graph from the adjacency matrix below



Q: Draw an adjacency matrix from the graph below





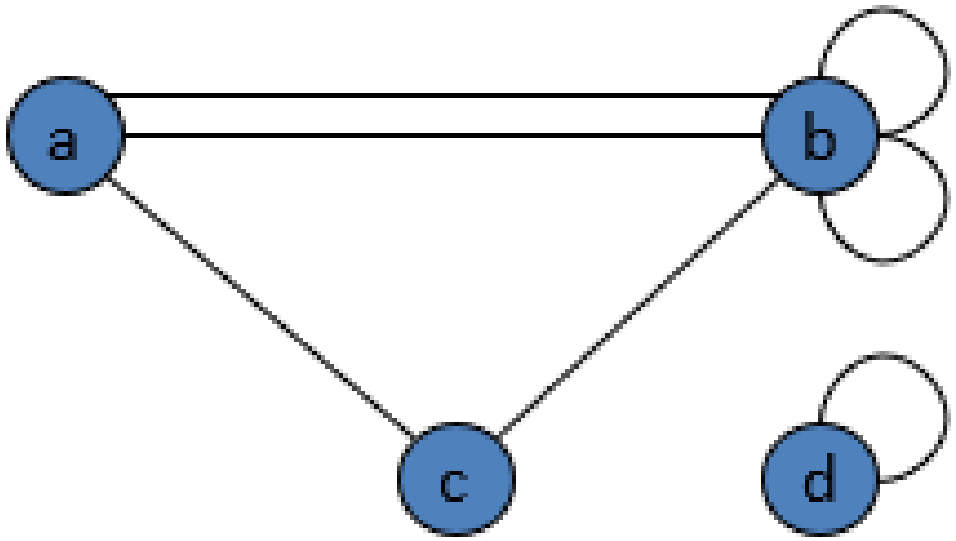


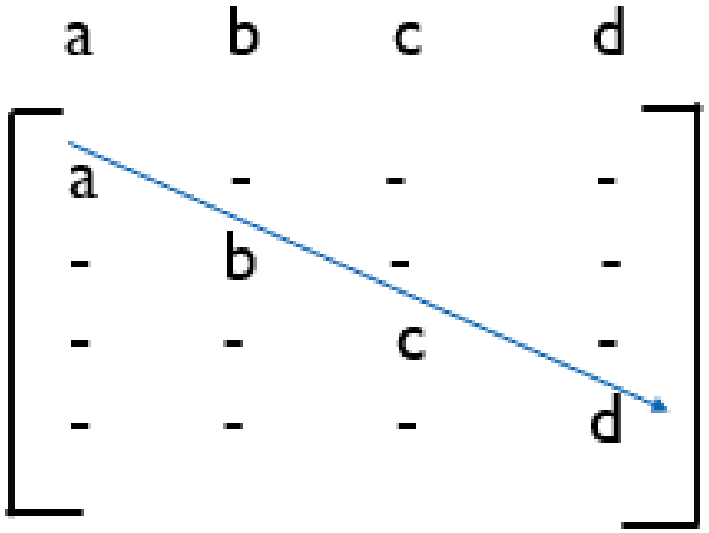
REPRESENTING GRAPHS: ADJACENCY MATRICES

- Suppose that G = (V, E) is a simple graph where |V|= n. Suppose that the vertices of G are listed arbitrarily as $v_1, v_2, \ldots, v_n$.
- The **adjacency matrix A** of G, with respect to this listing of the vertices, is the *n x n* zero-one matrix with 1 as its (i,j)th entry when v_i and v_j are adjacent, and 0 as its (i,j)th entry when they are not adjacent.
- If its adjacency matrix is **A** = [a_{ij}], then a_{ij} =1 if $\{v_i, v_j\}$ is an edge of G, 0 otherwise
- Adjacency matrix can also be used to represent **undirected graphs** with **loops** and with **multiple edges**.
- When multiple edges are present, the adjacency matrix is no longer zero-one matrix, because the (*i,j*)th entry of this matrix equals the number of edges that are associated to $\{a_i, a_j\}$
- A loop at the vertex a_i is represented by 1 at the (*i,j*)th position of the adjacency matrix.
- All **undirected graphs**, including multigraphs and pseudographs, have symmetric adjacency matrices (if A has edge with B then B also edge with A)

REPRESENTING GRAPHS: ADJACENCY MATRICES

Ex: What's the adjacency matrix of the following graph?





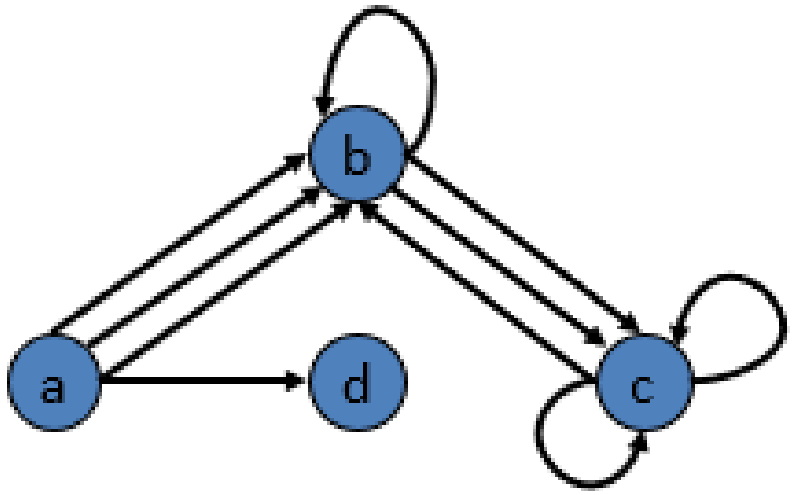
Solution:

$$\begin{pmatrix} 0 & 2 & 1 & 0 \\ 2 & 2 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

ADJACENCY MATRICES FOR DIRECTED GRAPH

- Adjacency matrices can also be used to represent **directed multigraphs**.
- However, such matrices are not zero-one matrices when there are multiple edges in the same direction connecting two vertices
- In the adjacency matrix for a directed graph, a_{ij} equals the number of edges that are associated to (v_i, v_j).

Q: what is the adjacency matrix of the following directed graph?

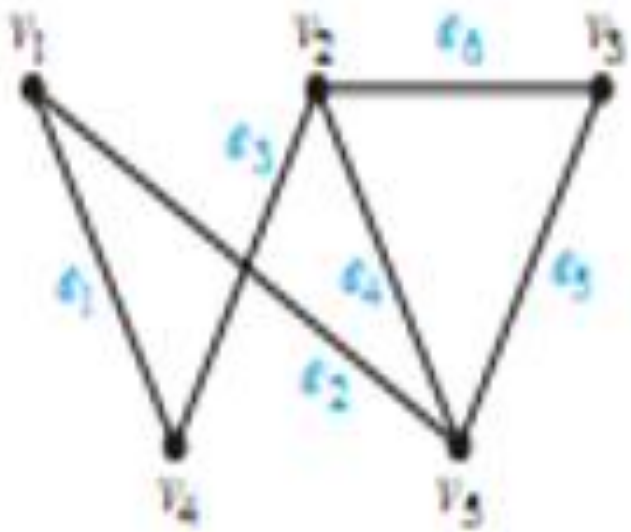


Solution:

$$\begin{pmatrix} 0 & 3 & 0 & 1 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

REPRESENTING GRAPHS: INCIDENCE MATRICES

- Incidence matrix: A matrix representing a graph using the incidence of edges and vertices.
- Let G = (V, E) be an undirected graph. Suppose that $v_1, v_2, \ldots, v_n$ are the vertices and $e_1, e_2, \ldots, e_m$ are the edges of G then the incidence matrix with respect to this ordering of V and E is the n x m matrix, $M = [m_{ij}]$, where $m_{ij} = 1$ when edge e_j is incident with v_i , 0 otherwise



$$
\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array}
\begin{array}{c}
\begin{array}{cccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{array} \\
\left[\begin{array}{cccccc}
1 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 1 & 0 & 1 \\
0 & 0 & 0 & 0 & 1 & 1 \\
1 & 0 & 1 & 0 & 0 & 0 \\
0 & 1 & 0 & 1 & 1 & 0
\end{array}\right]
\end{array}
.
$$

REPRESENTING GRAPHS: INCIDENCE MATRICES

EXAMPLE 7 Represent the pseudograph shown in Figure 7 using an incidence matrix.

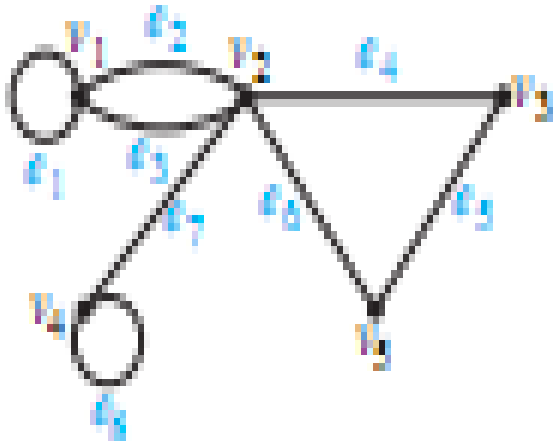


FIGURE 7 A Pseudograph.

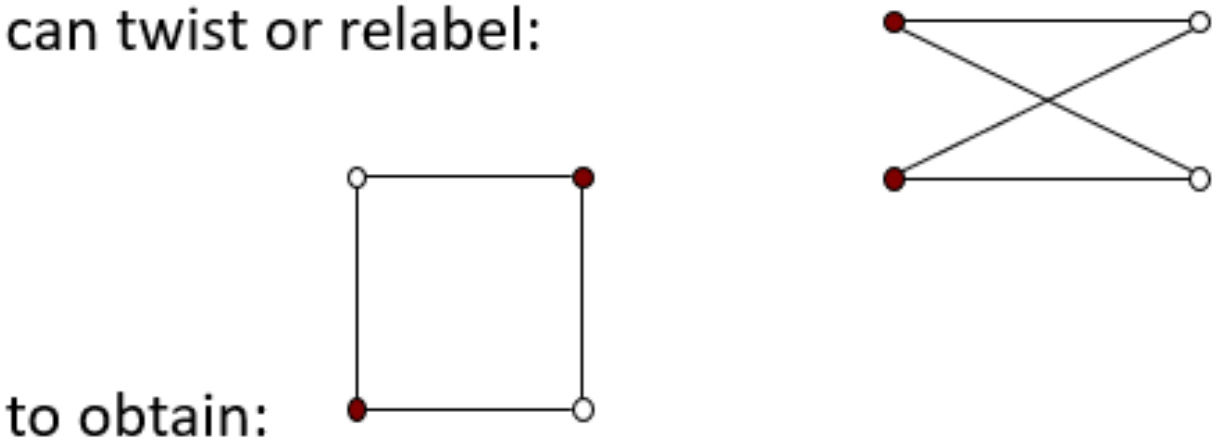
Solution: The incidence matrix for this graph is

$$\begin{array}{c} \\ v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{array} \begin{array}{c} \begin{array}{cccccccc} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 & e_8 \end{array} \\ \left[\begin{array}{cccccccc} 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{array}\right] \end{array}.$$

ISOMORPHISM OF GRAPHS

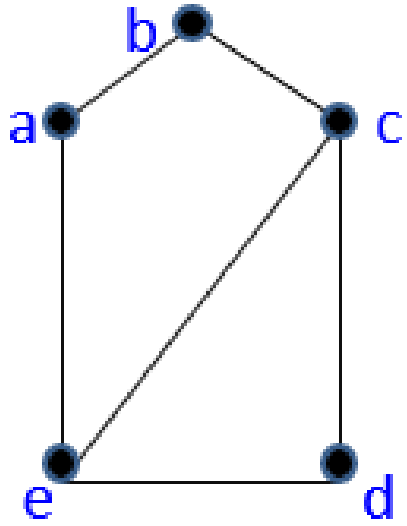
- The simple graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are isomorphic if there is a one-to-one and onto function f from V_1 to V_2 with the property that a and b are adjacent in G_1 if $f(a)$ and $f(b)$ are adjacent in G_1, for all a and b in V_1. Such a function f is called an isomorphism.
  - When two simple graphs are isomorphic, there is a one-to-one correspondence between vertices of the two graphs that preserves the adjacency relationship
  - Two graphs are isomorphic if they are identical except for their node names
  - Isomorphism of simple graphs is an equivalence relation

- Intuitively, two graphs are isomorphic if can bend, stretch and reposition vertices of the first graph, until the second graph is formed. Etymologically, *isomorphic* means "same shape".

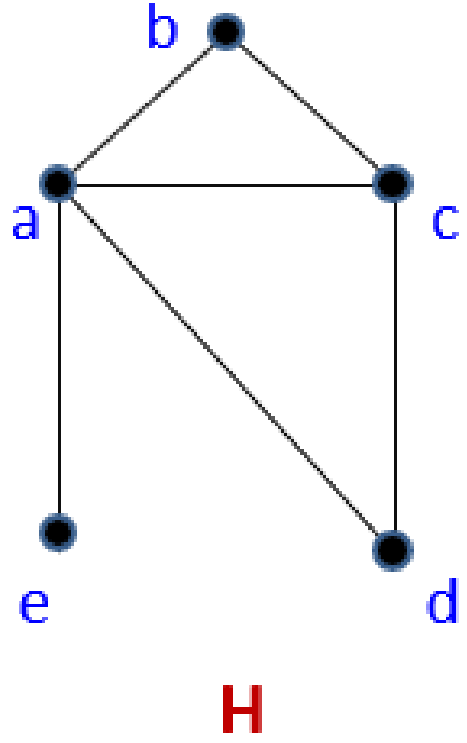


ISOMORPHISM OF GRAPHS

- It is often difficult to determine whether two simple graphs are isomorphic. Sometimes it is not hard to show that two graphs are **not isomorphic**.
- A property preserved by isomorphism of graph is called a ***graph invariant***.
- Isomorphic simple graphs must have –
  - The same # of vertices
  - The same # of edges
  - The degrees of the vertices must be the same
  - The same # and length of circuits (i.e. path starting from a-b-c-e-a is a circuit in graph G)



G



H

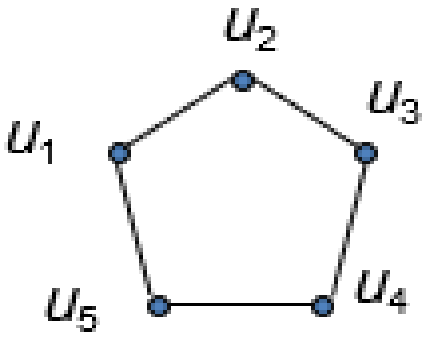
Q. Show that the following two graphs G and H are not isomorphic.

Solution : Both G and H have five vertices and six edges. However, H has a vertex of degree one, namely e, whereas G has no vertices of degree one. It follows that G and H are not isomorphic.

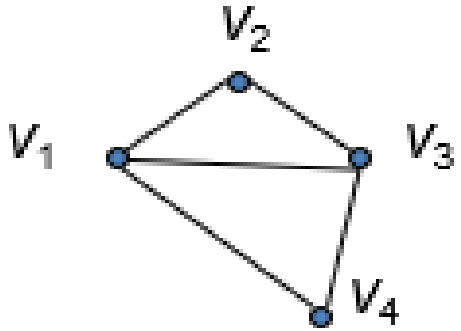
ISOMORPHISM OF GRAPHS

- **Note**: The number of vertices, the number of edges, and the number of vertices of each degree are all invariants under isomorphism. If any of these quantities differ in two simple graphs, these graphs cannot be isomorphic.
- However, when these invariants are the same, it does not necessarily mean that the two graphs are isomorphic. There are no useful sets of invariants currently known that can be used to determine whether simple graphs are isomorphic.

Q: why are the following graphs not isomorphic?



G

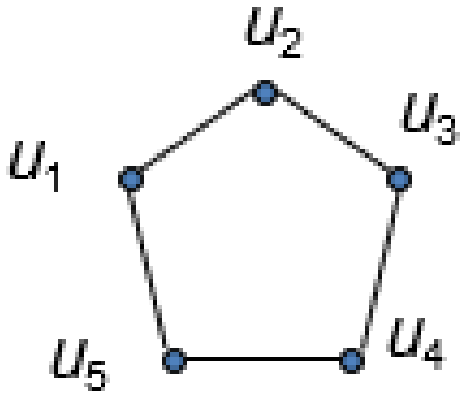


H

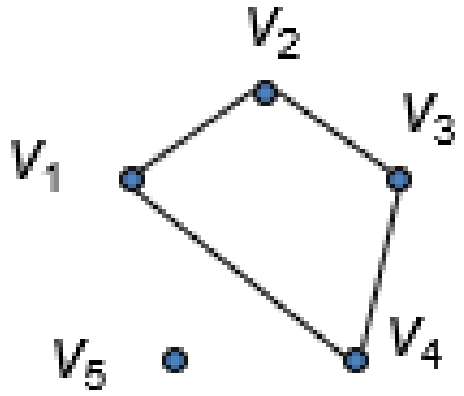
Solution: 1st graph (G) has more vertices than 2nd graph (H). So the graphs are not isomorphic

ISOMORPHISM OF GRAPHS

Q: why are the following graphs non-isomorphic?

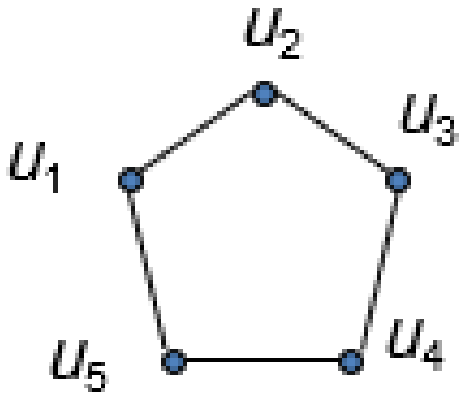


G

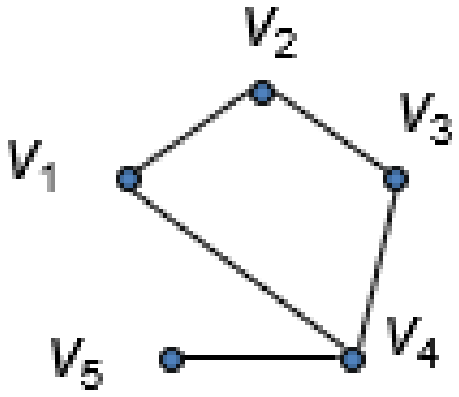


H

Solution: 1st graph (G) has more edges than 2nd graph (H). So the graphs are not isomorphic



G

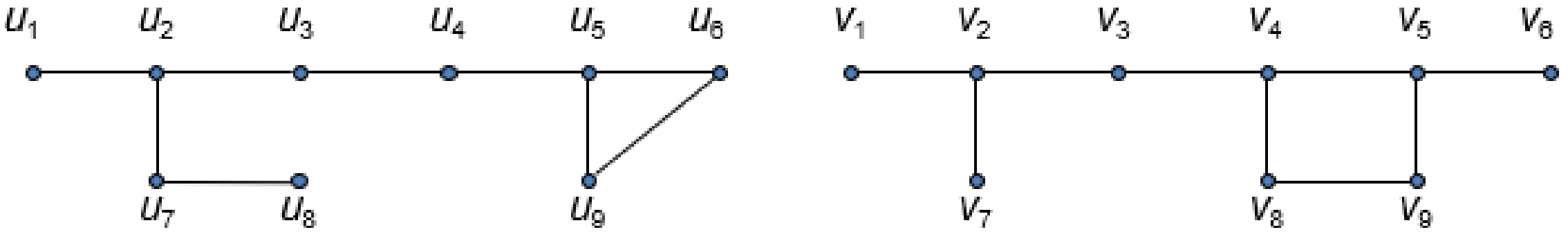


H

Solution: 2nd graph (H) has vertex of degree 1, where the1st graph (G) doesn't. So the graphs are not isomorphic.

ISOMORPHISM OF GRAPHS

Q: why are the following graphs non-isomorphic?



Solution: 1st graph (G) has 2 vertices of degree 1, whereas the 2nd graph (H) has 3 vertices of degree 1. So the graphs are not isomorphic.

REFERENCES

- *Discrete Mathematics and its applications with combinatorics and graph theory (7th edition)* by Kenneth H. Rosen [Indian Adaptation by KAMALA KRITHIVASAN], published by McGraw-Hill, is the required text.
- *SCHAUM'S ouTlines Discrete Mathematics(2nd edition)*, by Seymour Lipschutz, Marc Lipson
- *Discrete and Combinatorial Mathematics(3rd Edition)* by *R. P. Grimaldi,*, published by Addison-Wesley